Review on Aerobic Capacity and Anaerobic Power Demand of Fencers

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ABSTRACT

Fencing is a unique sport, which is reflected in the asymmetrical development of the muscles involved. Furthermore, the intermittent nature of fencing puts demands on both the aerobic and anaerobic metabolic systems. Many studies were conducted on Aerobic Capacity and Anaerobic Power on fencing. Review of previous studies observed that moderate aerobic power is necessary to support the sub-maximal changes of direction of the fencer to achieve the control of the game and to refuel the anaerobic mechanisms during the repeated interruptions occurring through the fencing bouts. The objectives of the present study were to review on aerobic fitness, anaerobic power parameters and their correlation with performance score. The data so obtained can serve as a baseline to be used by exercise physiologists, coaches, training academies to set training targets for their trainee fencers. Fencing, Aerobic Capacity, Anaerobic Power and Exercise Physiology

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INTRODUCTION

Fencing is a sport plays in Olympics in which the athletes combat against each other with a blade. It is simple and cheap to start and keeps both body and brain active. It is one of the oldest sports where the sword is used as shown by Egyptian frescoes circa (1200 BC) (Amberger, Johann Christoph–1999). In the year of 1896, it was introduced as the Olympic game held at Athens(Greece) (G.S. Roi, D. Bianchedi, 2008) and along with it in 1913 the Federation of International d'Escrime was formed (Amberger, Johann Christoph – 1999). In the year of 1974, F. A. I (Fencing Association of India), was founded and it was recognized in 1997 by the Government and affiliated to Indian Olympic Association, Asian Fencing confederation, Commonwealth Fencing Federation and Federation International D'Escrime. There are 6 individual and 16 events in 3 different weapons used in fencing - Epee, Foil, Sabre(Evangelista, Nick–1996). It is taken by Sports Authority of India as "Special Area Games" from 1989 to 1996.There are different type of energy systems that transform energy during game like Aerobic and anaerobic. Fencing is a physically and mentally challenging sport which involves offensive and defensive movements with

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a sword. The fencers' ability to act and react to opponent's sudden and unexpected movements is an important factor in success. In Fencing, a powerful lunge is key to a successful touch that counts towards the score. Power and velocity of Fencing lunge, opponents timing and actions, distance between fencers, strategy / intent of the opponent are other factors also affecting overall performance score.

Fencing is an intermittent sport that requires the fencers to compete in frequent, short bouts of high intensity exercise, followed by periods of low intensity activity. Bounces, steps of different direction and lunges occur repeatedly during the competition for the purpose of hitting the opponent, which also puts considerable demands on the neuromuscular system (Czajkowski, 2005).

It is an anaerobic sport that demands aerobic fitness to sustain the number of bouts required to be performed to reach the finals in the tournament. Since the tournaments are spread over the day consisting of multiple bouts, hence the relevance of assessing fencers for their endurance levels and including the same in their training schedules is important.

Studies have reported that although the aerobic capacity of fencers (52.9ml/Kg/min) is greater than sedentary population (42ml/Kg/min), it is lower than that of aerobic endurance athletes, therefore concluding that high VO2max has a relatively small role in Fencing.

Contrary to this, Stewart et al concluded that a high degree of cardio-respiratory fitness is a prerequisite for successful Fencing. They found that VO2 max, VE max & 2 Km run accounted for more of the explained variance in their fencing subjects. Anthony turner and his co-workers in their review article on Fencers have reiterated the importance of strength and conditioning training in Fencers Explosive power of lower limbs is important during attacks and counterattacks in a fencing event. Hence assessment of anaerobic test parameters and their monitoring is relevant for fencers. Researchers in the past have examined the usefulness of biomechanical and physiological indices recorded for estimation of adaptation levels. There is scarcity of data available on physiological profile of Indian elite fencers. Hence the present study endeavours to assess fitness profile of Indian elite fencers.

The objectives of the present study were to review on aerobic fitness, anaerobic power, other physiological profile parameters and their correlation with performance score. The data so obtained can serve as a baseline to be used by exercise physiologists, coaches, training academies to set targets/predict medal tally / podium positions for their trainee fencers.

LITERATURE REVIEW:

Many studies were conducted on Aerobic Capacity and Anaerobic Power on fencing. Review of previous studies observed that moderate aerobic power is necessary to support the sub-maximal changes of direction of the fencer to achieve the control of the game and to refuel the anaerobic mechanisms during the repeated interruptions occurring through the fencing bouts.

The absence of the important role of VO2max in fencing performance would be confirmed by the absence of a resting bradycardia that is a typical adaptation to chronic endurance exercise. With many completely different aerobic adaptations, these athletes experience a reduced endurance component in comparison to triathletes, combined with a high isometric component associated with fencing during the basic on-guard position, but this endurance component is always lower in individuals who practise fencing only.

Fencing requires a high level of anaerobic power for performance success. The motor response of a fencer is required to be fast and explosive. In the fencing competitions, the winner of the bout is the first of the two fencers to reach 5 heats in the pool round, and 15 heats in the instance of direct elimination. In different fencing events like Épée and Foil, there is a time limit of 3 minutes in the pool round and three 3-minute periods separated by 1-minute breaks in direct elimination. In these disciplines, if the bout ends before either fencer has reached 5 (or 15) hits, then the fencer with the most hits is declared the winner. In the modality of Sabre, there is a 1-minute break in the instance of direct elimination when one fencer's score reaches 8. Sabre often involves faster footwork and quicker bouts than Épée and Foil.

Fencing is a unique sport, which is reflected in the asymmetrical development of the muscles involved. Furthermore, the intermittent nature of fencing puts demands on both the aerobic and anaerobic metabolic systems. The Intermittent nature of fencing suggests that there is a high demand on the phosphorcreatine system, especially with a work: rest of 8s:10s.
A World Class fencer should therefore have developed both their aerobic and anaerobic capacities. Previous research has demonstrated the heart rate to be in the range of 167 to 191 beats•min-1 for 60% of the fencing duration during a Women's epee competition.

Maximal anaerobic alactic power (W) was studied by jumping on a force platform. In a sample of 11 top level épée fencers, W was 56.0±6.6 W/kg, with insignificant differences between control subjects and different levels of fencers 1241 A 20-second Wingate-type test performed by British épée fencers of international level showed a peak power of 809 + 38 W, which was not significantly higher during the in-season compared with the off-season. Even the correlations between dominant/non-dominant leg.

This study suggests that the performance of the fencers requires a good aerobic capacity and the success in fencing depend on cardio-respiratory fitness parameters. Furthermore, there was no significant correlation between the fencing scores and the 1 min (Manahan and Gutin 1971) step test. Since fencing includes many short, bursts of high intensity activity, it probably taxes anaerobic energy sources to a large extent.

The non-significant correlations were probably attributable to the step-test results being inadequate for indicating anaerobic capacity Stewart et al., 1977 reported that high state of cardio-respiratory fitness was required for being successful fencer. They found a positive correlation between fencing scores and aerobic capacity. In this study for VO2 max testing the subjects ran at 6mph beginning the test at 0% grade and this was increased by 0.5% every 2 mins until the subject no longer continue, the highest VO2 observed was considered as VO2 max.

A study was conducted by J. Nyström et.al. (1990) on 6 Swedish epee players to perform treadmill test for maximal oxygen uptake. The speed was set to 14 km/h with an inclination of and the inclination was increased with 1° every minute following that point. A blood lactate sample was obtained within 30 sec following exhaustion. It was observed that the maximal oxygen uptake is 67.3 ± 3.7 ml/kg/min. The result showed fencers have high maximal aerobic power.

Seven international British players were taken for the study of physiological parameters by Y. Koutedakis et.al. (1993).

The time protocol for wingate type test was 20s and the resistance was 8% of subject's body weight. The mean VO2 max is lower in in-season period (54.8ml/kg/min) as compared to off-season period (58ml/kg/min). There was no statistical difference observed in Wingate test (mean power, peak power).So, it can be concluded with that, VO2 max demonstrates that a significant seasonal variation, and there was occurrence of sporting success if there was aerobic activity lasting over 2 mins and for the anaerobic power the competition itself is sufficient to maintain the power throughout the in-season competition period.

In this case it was assumed that all aerobic energy derived from carbohydrate oxidation. Blood samples were obtained with a finger prick and blood lactate concentration was measured at rest, at the end of each round and at the 3rd and 5th 134 min of the final recovery by using a portable lactate analyzer (Lactate Pro, Arkray Inc., Kyoto, Japan), to evaluate the time course of lactate concentration.

Cerizza C, Roi G (1994) studied that defining the predominant anaerobic system, originates from reports quantifying the blood lactate concentrations of fencing bouts. In men's foil, for example, blood lactate concentrations (measured 5 minutes after bout) averaged 2.5 mmol/L during the preliminary pools and then were consistently above 4 mmol/L (and as high as 15.3 mmol/L in the winner) during the elimination bouts.

Borisiuk Z. (2005) reported that the saber fencers are facing considerable requirements pertaining to releasing and rebuilding anaerobic energy resources. This can be confirmed by the high parameters of anaerobic power. Therefore, optimal preparation of saber fencers should take into account long-time efficiency training. As the study showed, the mean VO2max of saber fencers amounted to 57.96 ml/kg/min. Saber fencing training should focus on development of speed predispositions on the basis of overall endurance ,Roi and Bianchedi (2008).

Roi and Bianchedi (2008) reported that although the aerobic capacity of fencers (52.9 mL/kg per minute) is greater than that of the sedentary population (approximately 42 mL/kg per minute), it is clearly lower than that of aerobic endurance–based athletes and may be suggestive of the relatively small role a high VO2max has to fencing.

Ratmess (2008) found that fencing relies predominately on anaerobic metabolism; aerobic energy system contribution may be small and predominately involved in the sub maximal movements of the "on guard" position and during recovery periods (inter bout and intra bout). Furthermore, although the energy system requirements of each sword will inevitably differ, it is in the opinion of the authors that none will significantly tax the aerobic system to the extent that training need directly target its development through the traditional methods of long slow distance (LSD) running.

Bottoms M et.al. (2011) suggested that fencing training should focus on developing both aerobic and anaerobic (Creatine Phosphate and Alactic) metabolism. Introducing specific training that simulates the demands of a competitive bout could help develop the necessary fitness for competitive fencing. Given the high intensity nature of fencing, it is important that athletes replenish the energy lost during competition to ensure that performance is not adversely affected.

According to Milia R etal.,(2014) both the aerobic energy metabolism and anaerobic lactic energy sources are important and moderately recruited. Thus, present data support the concept that successful fencing performance depends more on skill and technique than the superior aerobic and anaerobic capacities.

Turner, Anthony (2013) found that although foil fencing is undoubtedly an anaerobic- type sport, it appears that the preliminary bouts rely more on the alactic system, whereas the elimination bouts rely more on the lactate system.

Oates, L.W. et.al. conducted a study (2019) on 8 elite male epee fencers to determine the physiological demands of epee fencing performance. Portable breath by breath gas analysis system (Cosmed K4b2) is used to record expired gas data on an average over 5 sec period during the time of combat to calculate average and maximum oxygen consumption. The result showed similar mean and maximum VO2 in case of both poule and direct elimination fights and they concluded with a suggestion that to deal with the physiological demand of the game the fencers must acquire a high level of aerobic fitness

Bhatt. M et.al. (2021)reported that fencing score was found to have weak and statistically insignificant positive correlation with relative peak power. Pearsons correlation between VO2max and Fencing score was found to be weakly positive but statistically insignificant respectively. In this study Lower body Wingate test was performed using Monark 894e Lower body cycle ergometer.

CONCLUSIONS:

It is observed after Review of literature related to the physical demands of fencing competitions are high, involving the aerobic and anaerobic alactic and lactic metabolisms, and are also affected by age, sex, level of training and technical and tactical models utilized in relation to the adversary.

Fencing produces typical functional asymmetries that emphasize the very high level of specific function, strength and control required in this sport. Moreover, the physical demands of fencing are closely linked to the perceptual and psychological ones, and all are subjected to a continuous succession of changes during the bouts based on the behaviour of the opponent. Competitive fencing is a varied and challenging sport. Therefore, it is not surprising that one of the best determinants for success in Fencing is explosive anaerobic power. However high level of aerobic power plays a significant role in muscular strength, muscular power. Thus, aerobic power testing of a Fencer is as equal important as determining the anaerobic power.

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